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Chouaib Doukkali University
National School of Applied Sciences of El Jadida
Master Data Science and Artificial Intelligence
Module: Natural Language Processing

ENSA
EL JADIDA

IMDb Movie Review Sentiment Analysis

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Presentation outline

01 Introduction and problem

02 Dataset and acquisition

03 Preprocessing and representation

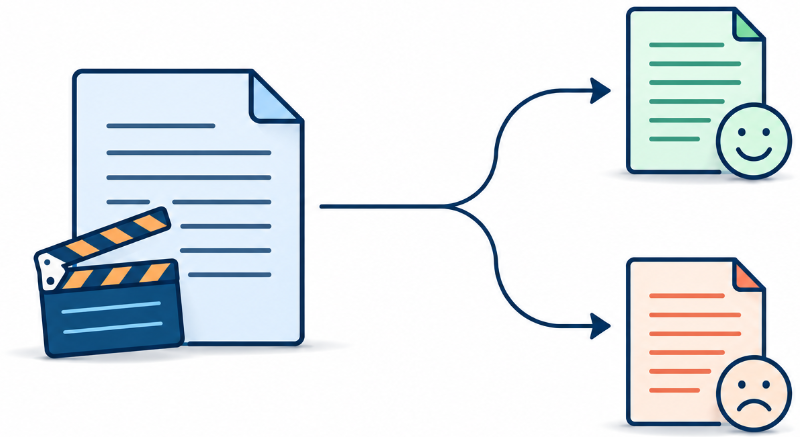
04 Modeling: SVM and CNN

05 Evaluation and critical analysis

06 Application and demonstration

07 Conclusion and future work

Binary sentiment classification



Each review receives one sentiment label

Dataset and experimental split

50,000 reviews

English movie reviews
25,000 positive / 25,000 negative



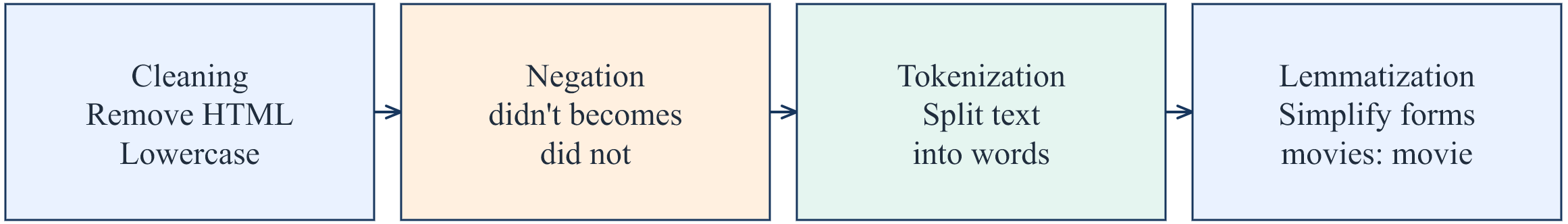
418 exact duplicates + 4 after cleaning
49,578 unique reviews remain
No missing or blank reviews

Split	Reviews	Purpose
Train	39,662	Fit models
Validation	4,958	Select checkpoint
Test	4,958	Final evaluation

80% / 10% / 10%

Stratified by sentiment, random seed 42

Preprocessing



Original THE movies weren't boring.

Processed **the movie be not boring**

Keep negation: "good" and "not good" have different meanings.

Text representation

TF-IDF

Weights for words and word pairs

50,000 features
Individual words and pairs of words
Example: "not recommend"

Efficient representation for SVM

GloVe embeddings

Dense word vectors with 100 values

Up to 20,000 vocabulary entries
Maximum 450 tokens per review
100 numbers per word

Pretrained word information for CNN

Both vocabularies use training reviews only

TF-IDF with SVM

Support Vector Machine

Input: TF-IDF features
Output: positive or negative
Model: linear SVM

Separates positive and negative reviews

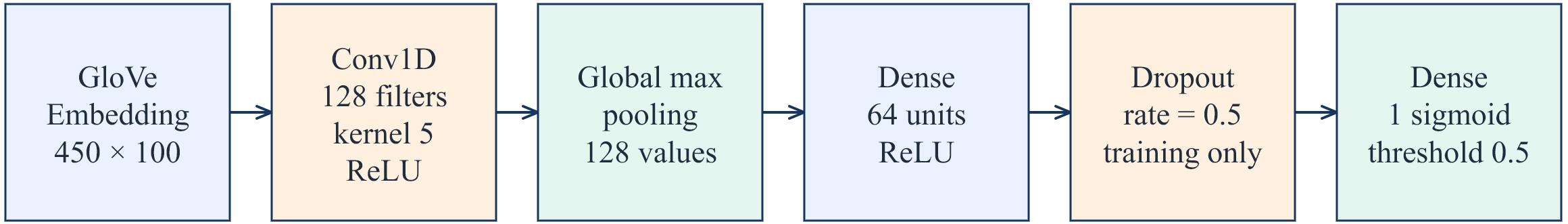
Efficient for text classification
Word weights help explain predictions

Feature	Learned weight
excellent	+3.42
great	+3.23
awful	-4.60
bad	-4.58
not recommend	-3.53

Positive weights favor positive sentiment.
Negative weights favor negative sentiment.

GloVe with CNN

450 tokens per review

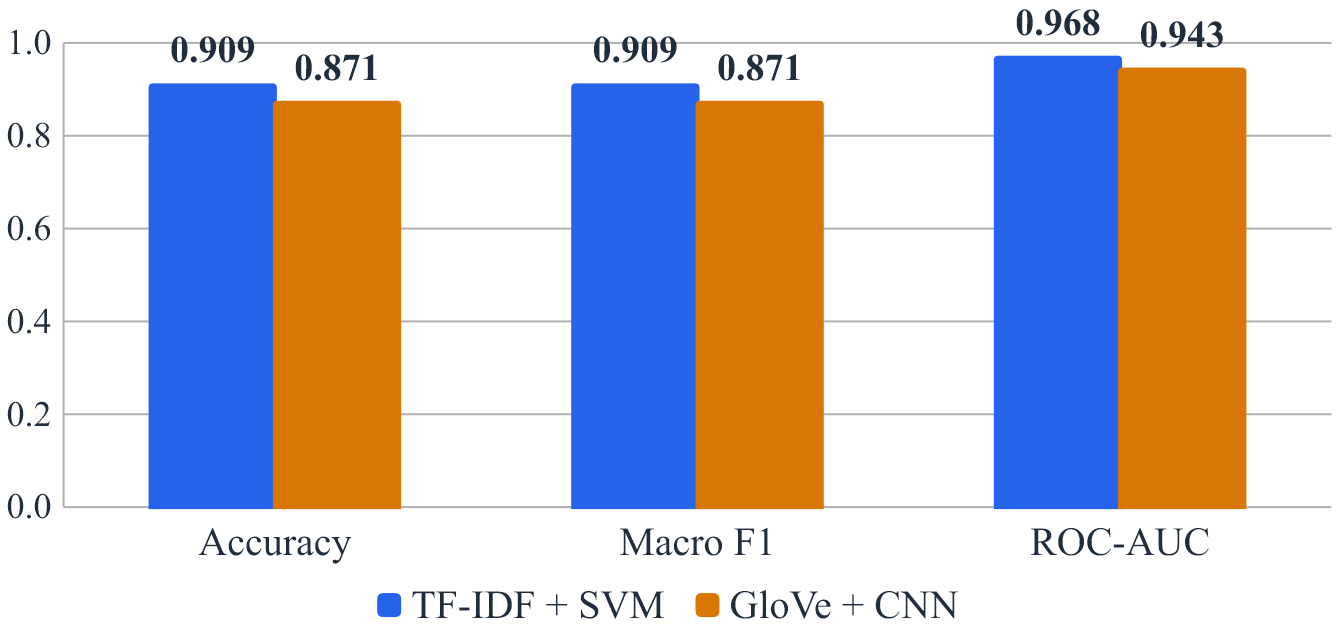


Pattern detector

Looks at five tokens at a time
Combines patterns across the review

Dropout: 0.5
Early stopping on validation loss
Frozen GloVe embeddings

Test results



Metric	SVM	CNN
Accuracy	0.9086	0.8711
Macro precision	0.9089	0.8711
Macro recall	0.9086	0.8711
Macro F1	0.9086	0.8711
ROC-AUC	0.9682	0.9426

90.86% vs 87.11% accuracy

4,958 test reviews. SVM leads by 3.75 percentage points for these configurations.

Confusion matrices

TF-IDF + SVM

Predicted label

Actual label	Negative	Positive
Negative	2211	259
Positive	194	2294

453 errors

GloVe + CNN

Predicted label

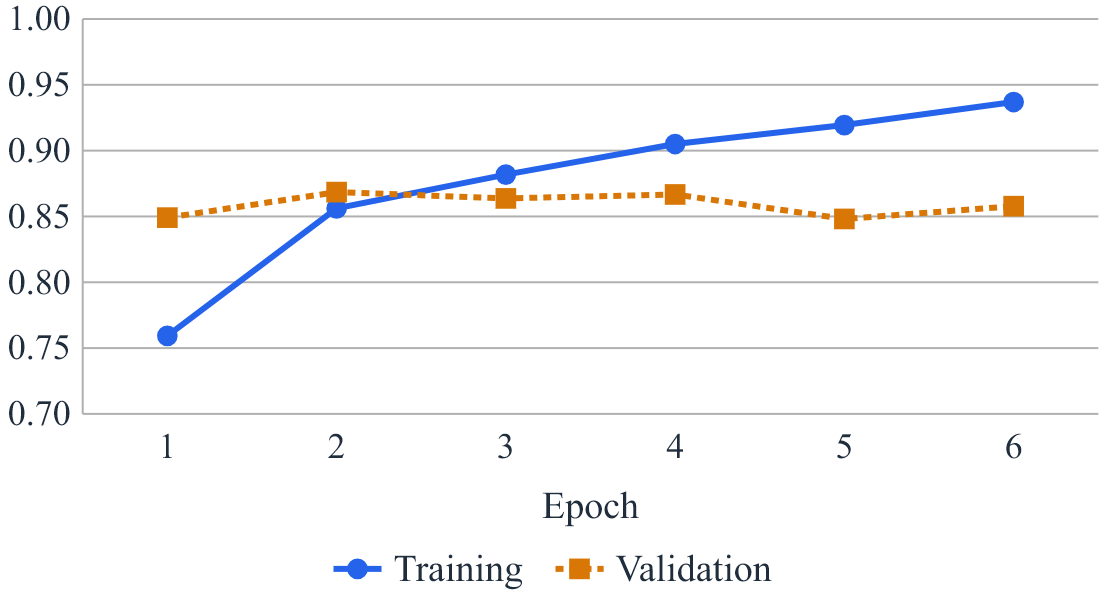
Actual label	Negative	Positive
Negative	2142	328
Positive	311	2177

639 errors

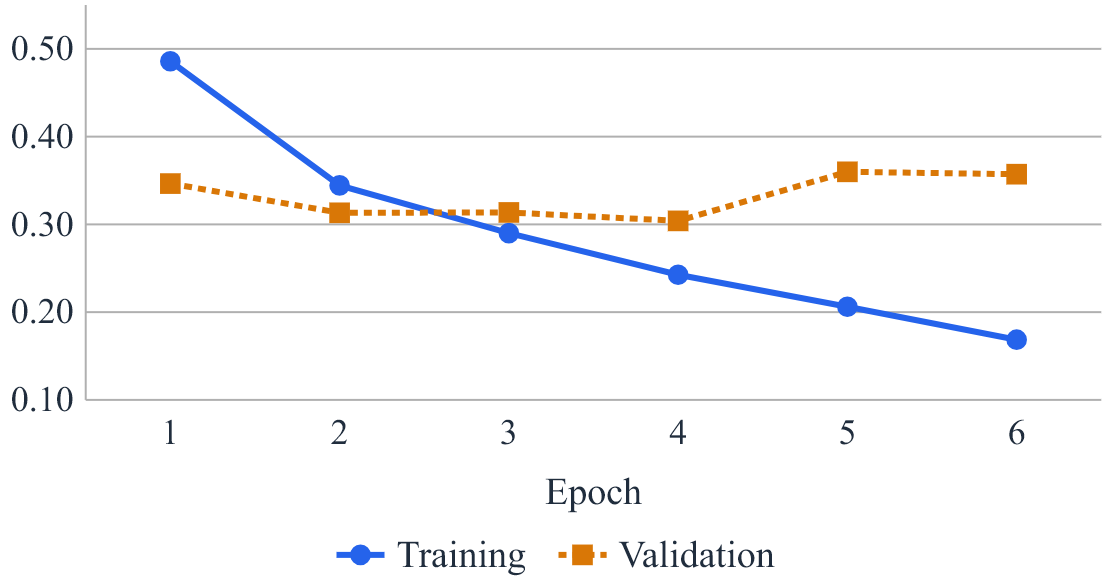
False positive: negative predicted as positive. False negative: positive predicted as negative.

CNN overfitting

Accuracy



Binary cross-entropy loss



Selected checkpoint: epoch 4

Minimum validation loss = 0.3040. Training stops at epoch 6 and restores the best weights.

Error analysis and limitations

“The film was decent, but I didn’t like the way it ended;
I believe there should have been a different, more logical conclusion.”

SVM: Positive

Mixed opinions

Praise: "The film was decent"
Criticism: "I didn't like the way it ended"

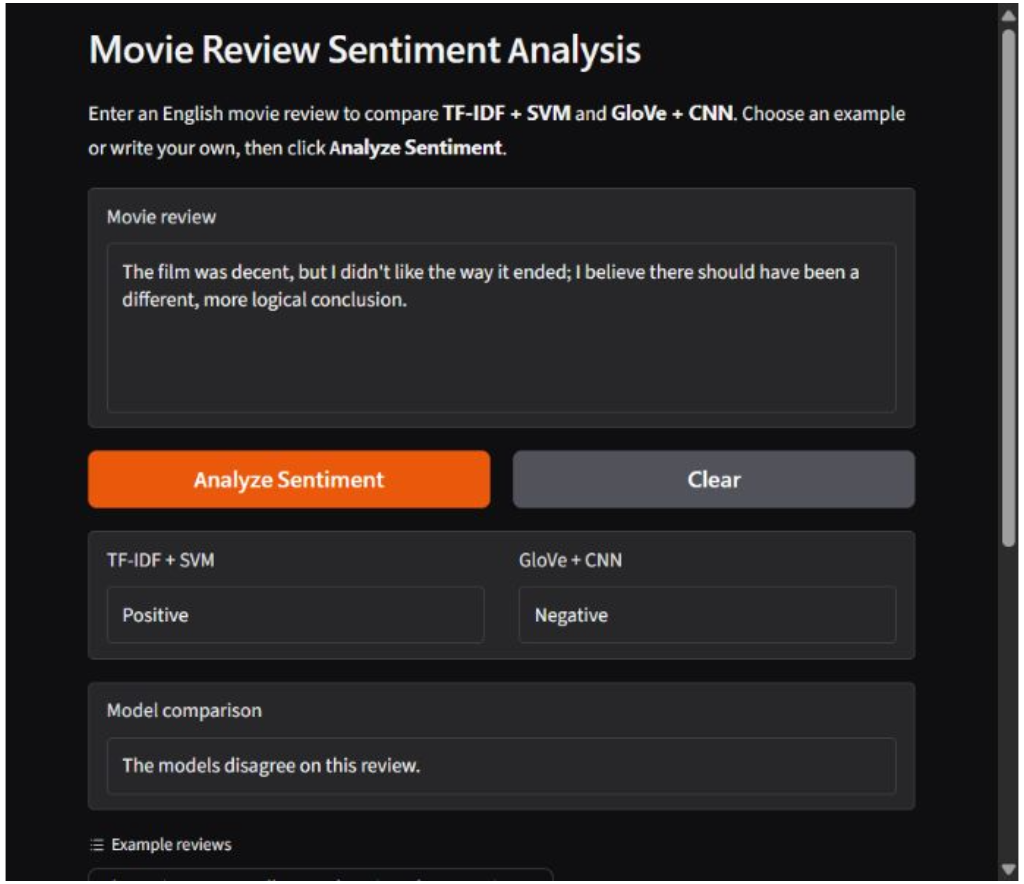
CNN: Negative

Limits of the experiment

English movie reviews only
Two sentiment classes
More testing needed for generalization

No official label for this example. Disagreement alone does not tell us which model is correct.

Application demonstration



Live demo

1 Enter an English review
2 Click Analyze Sentiment
3 Compare both predictions

Train on Kaggle
Predict locally on the CPU

Saved models and identical preprocessing
Original Keras weights via NumPy

Conclusion and future work

TF-IDF + SVM is the stronger model in this experiment

90.86% test accuracy with a working local demo

Future work

Improve the CNN
Test different configurations using validation data
Evaluate on other datasets

Thank you
for your attention

Questions?

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